

Challenge (Habanero)

- Q1.** Factor out the greatest common factor from $12x + 18y$
(A) $6(2x + 3y)$
(B) $3(4x + 6y)$
(C) $2(6x + 9y)$
(D) $9(4x/3 + 2y)$
(E) $4(3x + 9y/2)$
- Q2.** A construction worker needs to transport 15 boxes of tiles, with each box containing 8 tiles. What is the total number of tiles, factored by the greatest common factor?
(A) $8 \cdot 15$
(B) $5 \cdot 24$
(C) $3 \cdot 40$
(D) 120
(E) $15 \cdot 8$
- Q3.** A recipe for cookies calls for 9 cups of flour and 12 cups of sugar. What is the greatest common factor of these two ingredients?
(A) 1
(B) 3
(C) 6
(D) 9
(E) 12
- Q4.** Simplify the expression $20x + 16y$ by factoring out the greatest common factor
(A) $4(5x + 4y)$
(B) $8(5x/2 + 2y)$
(C) $2(10x + 8y)$
(D) $10(2x + 8y/5)$
(E) $16(5x/4 + y)$
- Q5.** A builder needs to lay 24 bricks per row, with 8 rows in total. What is the total number of bricks, factored by the greatest common factor?
(A) $8 \cdot 24$
(B) $3 \cdot 64$
(C) $4 \cdot 48$
(D) 192
(E) $8 \cdot 3 \cdot 8$
- Q6.** A chef needs to prepare 18 servings of a dish, with 4 dishes per serving. What is the total number of dishes, factored by the greatest common factor?
(A) $2 \cdot 36$
(B) $3 \cdot 24$
(C) $6 \cdot 12$
(D) $9 \cdot 8$
(E) $18 \cdot 4$
- Q7.** Simplify the expression $50x - 25y$ by factoring out the greatest common factor
(A) $25(2x - y)$
(B) $5(10x - 5y)$
(C) $10(5x - 2.5y)$
(D) $25(2x - y)$
(E) $50(x - 0.5y)$
- Q8.** A contractor needs to install 36 windows, with 6 windows per row. What is the total number of rows, factored by the greatest common factor?
(A) 6
(B) $6 \cdot 6$
(C) $3 \cdot 12$
(D) $2 \cdot 18$
(E) $9 \cdot 4$

Open Ended Questions (FRQ)

- Q9.** Factor out the greatest common factor from the expression $24x + 30y - 18z$ and simplify.
- Q10.** A recipe calls for 12 cups of flour and 15 cups of sugar. If the chef wants to quadruple the recipe, what is the total number of cups of flour and sugar needed, factored by the greatest common factor?

Chef's Tips

- Check for the greatest common factor in each question
- Ensure students simplify expressions correctly
- Emphasize the importance of factoring in real-world applications like cooking and construction